LifeXA: An AI Assistant for Everyday Task Management

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***Abstract*—Personal task and reminder tools usually treat every task and every user in nearly the same way, leaving the user to decide what to do, when to do it and how to prioritise it. This paper presents LifeXA, a psychologically informed AI personal assistant that combines a Large Language Model (LLM) interface with user-controlled memory, adaptive reminder logic, voice interaction and selectable communication personalities. LifeXA is task-oriented: it is not a therapist, a diagnostic system or an emotional companion. We describe the psychological foundation of the design (cognitive load, goal setting, procrastination, motivation, routine formation and personalisation), a five-level personalisation model, and a hybrid architecture in which deterministic rule-based logic handles scheduling, permissions and reminders while the LLM handles language. We then outline a staged research methodology and an evaluation plan that compares a conventional assistant with the personalised LifeXA assistant on task management, perceived usefulness, engagement, trust and user autonomy. This paper reports the concept, design and research plan; empirical results are not yet available.**

***Index Terms*—AI personal assistant, large language model, personalisation, adaptive reminders, voice assistant, human–computer interaction, user autonomy, task management.**

I. INTRODUCTION

People manage personal tasks, college work, projects, meetings, appointments, reminders and routines through several different tools. Even when separate productivity applications are available, the user still has to decide what to do, when to do it, how to prioritise it and which reminder is actually useful. A conventional reminder treats tasks in almost the same way: a notification may simply state that an assignment is due at 7 PM. It does not know whether the user normally studies at night, whether the task has already been postponed, whether another important activity overlaps, or whether a shorter task breakdown would be more helpful.

LifeXA is an AI-based personal assistant that helps a person manage everyday activities in one place. It is not intended to be another chatbot that only answers questions; its purpose is to help the user organise, remember, prioritise and act on the things that matter in daily personal and professional life. We use the following working definition: *LifeXA is a psychologically informed AI personal assistant designed to support daily functioning through personalised scheduling, task management, reminders and adaptive assistance.*

The psychology component concerns everyday behaviour—planning, remembering, prioritising, motivation, routines, procrastination and task completion. It is not clinical. The way an assistant gives reminders, breaks down tasks, adapts to routines and leaves the user in control affects how useful and acceptable it feels. An LLM supplies the natural-language layer, and the system is planned to support voice interaction and a user-selectable communication style.

The main contributions of this work are: (i) a framework for psychologically informed personal assistance; (ii) a hybrid LLM and rule-based architecture; (iii) a five-level personalisation model; (iv) adaptive reminder logic that preserves user autonomy; and (v) an evaluation plan for usefulness, engagement, trust and autonomy. Table I clarifies the intended role of the system.

TABLE I
WHAT LIFEXA IS AND IS NOT

LifeXA is	LifeXA is not
A personal assistant	A replacement for a psychologist
Task-oriented	An AI friend or emotional companion
Psychologically informed	A diagnostic mental-health system
Personalised and adaptive	Designed to create emotional dependence
User-controlled	A system making important decisions for the user
An LLM-enabled interface	A claim of clinical effectiveness

The scope covers personal planning (schedules, routines, to-do items, appointments, shopping, study routines), academic and professional work (assignments, exams, projects, internship tasks, meetings), and interaction through text, natural-language task creation, voice and selectable personality. Diagnosis, therapy, high-stakes decisions made on the user's behalf, manipulation to retain users and the creation of emotional dependence are explicitly out of scope.

II. RELATED WORK

A. Personalisation in Digital Interventions

Personalisation can be described as purposeful variation in content, structure, guidance or communication between users. A systematic review of 94 digital mental-health interventions found personalisation in 66% of them, with decision rules and user choice among the most common mechanisms; the review also noted that strong evidence for the benefits of personalisation was still limited and that machine-learning-based personalisation was uncommon [1]. A 2025 systematic review of personalised mobile apps for non-clinical populations likewise reported promising but preliminary evidence, with limitations such as small samples, attrition and few direct comparisons between personalised and non-personalised versions [2]. Evidence from mental-health applications cannot

automatically be treated as evidence for a general personal assistant.

B. Memory in Personalised LLM Assistants

Research on personalised LLM assistants has explored storing information from previous conversations and retrieving relevant memories so that later responses better match user preferences [3]. LifeXA adopts a deliberately practical form of memory, limited to information that is useful for task support and under the user's control.

C. Voice Assistants, Personality and Trust

Studies of voice assistants report that voice similarity and personality customisation can influence attraction, credibility and trust-related perceptions [4]. Human-like linguistic and behavioural traits have been linked with enjoyment, trust and willingness to continue using a voice assistant [5], and anthropomorphism and relationship-like perceptions can shape trust and interaction [6]. These findings motivate both the voice and personality features of LifeXA and the clear role boundaries described in Section VII.

D. Research Gap

These areas are often studied separately. We do not claim that no earlier system has combined them; the more realistic position is that the evidence is fragmented. LifeXA offers an opportunity to study how psychological principles, personalisation, adaptive reminders, LLM interaction, voice, selectable personality and user-controlled memory work together in an everyday, non-clinical, task-oriented personal assistant.

III. PROBLEM STATEMENT AND RESEARCH QUESTIONS

Storing tasks and sending notifications is not enough. An adaptive assistant can use preferences, routines, context and previous interactions to decide how information should be presented. The research question is therefore not simply whether AI can answer questions, but whether psychologically informed personalisation can make an AI assistant more useful for everyday task management.

Main research question. How can an AI-powered personal assistant use psychological principles, user preferences, behavioural patterns and contextual information to provide personalised support for daily personal and professional activities?

Focused research question. Does psychologically informed personalisation improve users' task management, engagement and perceived usefulness compared with a conventional, non-personalised personal assistant?

Supporting questions address how reminder timing, frequency and wording influence task completion; which types of user information are actually useful; how routines and behaviour can adapt assistance; whether voice interaction and selectable personalities change perceived usefulness, comfort and ease of use; how autonomy is preserved during proactive suggestions; how memory is stored, retrieved, updated and deleted; and how privacy, transparency and trust are maintained.

The research objectives are: (i) to investigate how psychological principles can be incorporated into an everyday-life assistant; (ii) to study how personalisation based on preferences, routines and behaviour improves relevance; (iii) to investigate adaptive reminders and task recommendations; (iv) to examine how an LLM can provide natural-language interaction while the system remains task-oriented and user-controlled; (v) to study how voice output and selectable personalities influence usefulness, comfort, engagement and trust; (vi) to examine user acceptance of the assistant; and (vii) to identify boundaries for privacy, autonomy, transparency and safe use.

IV. PSYCHOLOGICAL FOUNDATION

The psychology component focuses on practical concepts relevant to everyday functioning. These concepts guide how the assistant organises information and supports behaviour; they are not used to diagnose the user.

A. Cognitive Load and External Memory

People have limited attention for tracking several things at once. The assistant acts as external support for tasks, deadlines and plans. Instead of expecting the user to remember that an assignment is due tomorrow, LifeXA keeps the deadline, preparation time and preferred study period together and offers a timely suggestion.

B. Goal Setting and Task Breakdown

Large tasks can be hard to start when seen as a single activity. LifeXA can represent a project as research, data collection, analysis, documentation and final review rather than one large deadline.

C. Procrastination and Task Initiation

Repeating the same reminder will not necessarily solve procrastination. If a task is repeatedly postponed, the system should offer a smaller starting action, a different time or a rescheduling option instead of sending the same notification again.

D. Motivation

Motivation varies with task, time and situation. LifeXA therefore uses practical, non-manipulative support such as showing progress, suggesting a realistic next step or helping the user choose a suitable time.

E. Routine and Habit Formation

Repeated behaviour creates patterns. A regular study period, exercise days or a preference for planning at night become useful context. The pattern should support the user, not control the user.

V. PROPOSED SYSTEM

A. Layered Architecture

LifeXA is organised into seven layers (Fig. 1). The user interface provides chat, home screen, memory, profile, tasks and reminders. The voice layer handles speech input and

spoken responses. The LLM layer performs natural-language understanding and response generation. The memory layer holds relevant preferences, routines and saved context. The psychological/decision layer contains rules for prioritisation, timing, adaptation and user autonomy. The task and schedule layer manages tasks, deadlines, appointments and reminders, and the data/storage layer keeps user data, settings and system records.

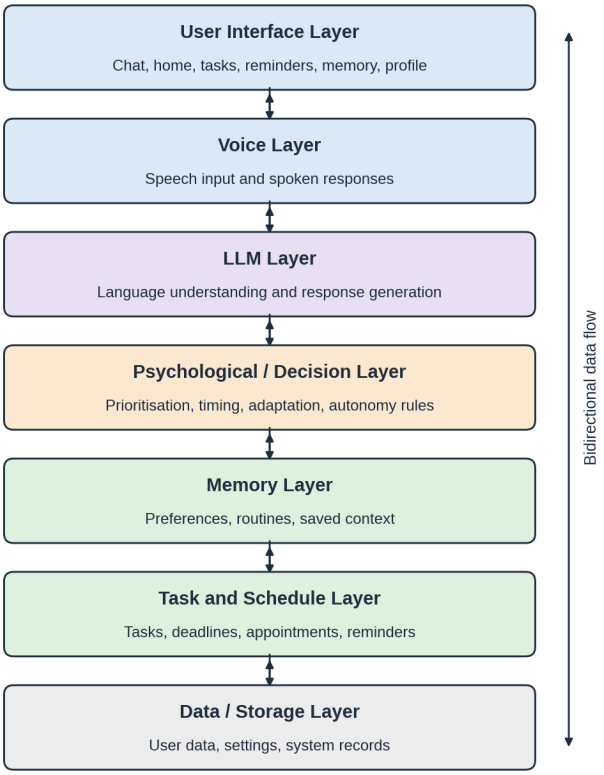


Fig. 1. Layered architecture of the proposed LifeXA system.

B. Hybrid LLM and Rule-Based Processing

The LLM is an important part of LifeXA but not the whole system. It converts conversational requests into structured tasks, answers questions, generates personalised wording, summarises plans, helps break larger tasks into steps and produces responses matching the selected personality. Task storage, reminders, permissions and scheduling are handled by deterministic application logic rather than by generated text.

Fig. 2 shows the intended flow: a voice or text request is understood by the LLM and converted into a structured intent; rules combine this with memory and schedule context to produce an action or recommendation; the LLM then phrases the reply in the chosen personality for text or voice output. A meaningful schedule change is never applied silently; the user is asked to confirm. This separation exists because the LLM is flexible with language but can be unpredictable, rules suit fixed behaviour, memory provides continuity, the decision layer carries the research component, and separate parts are easier to evaluate.

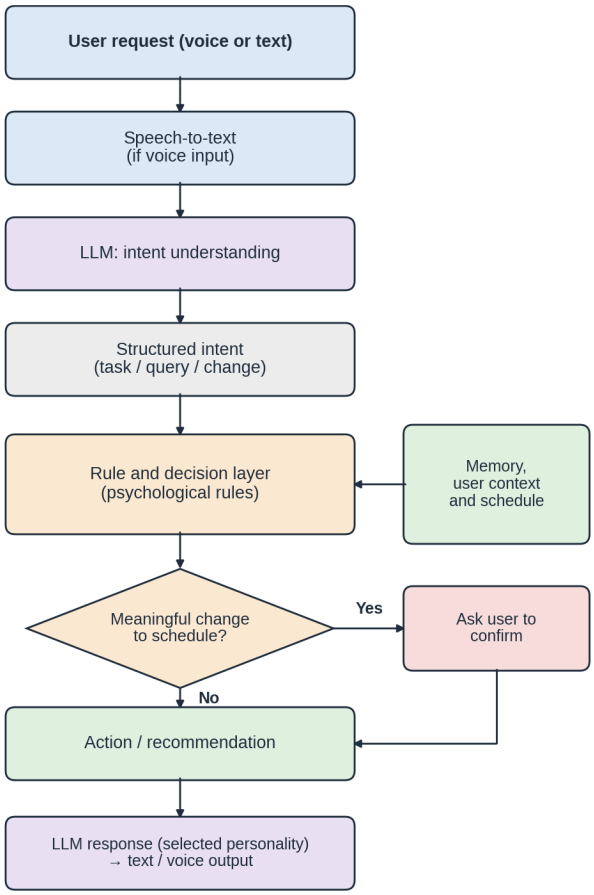


Fig. 2. Hybrid processing flow combining LLM understanding with rule-based decisions and user confirmation.

C. Personalisation Model

Personalisation is introduced in five levels (Fig. 3), so that the system need not collect every possible piece of information from the start. It begins with explicit user preferences, then uses repeated routine patterns, observed behaviour such as postponing or completing tasks, and current context such as a class schedule, and finally combines them as adaptive assistance. For example, the system may suggest a 45-minute study block after 8 PM.

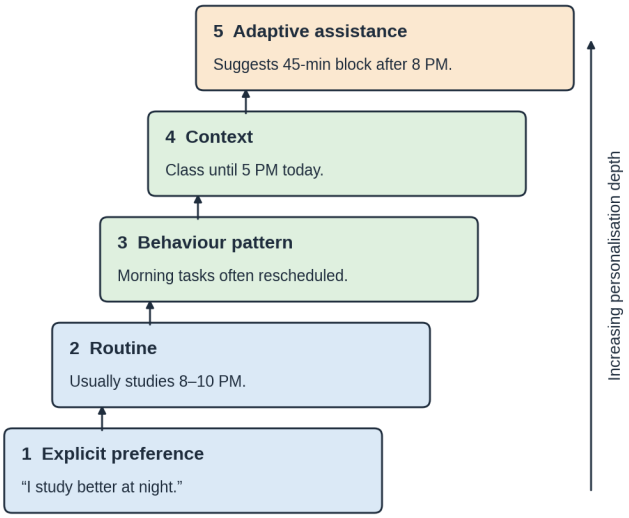


Fig. 3. Five-level personalisation model of LifeXA.

Personalisation remains user-controlled: users can edit or remove stored preferences and override suggestions; the assistant explains important assumptions behind a recommendation; learned patterns are not assumed to be always correct; sensitive information is not collected merely because it could be; and personalisation should improve usefulness without creating dependence.

Implementation status. At the time of writing, the prototype implements the memory layer (Level 1 explicit-preference personalisation), the LLM layer, and user-controlled data storage with full create, read, and delete access; the remaining architectural layers — voice interaction, adaptive reminder logic, selectable personality, and Levels 2–5 of the personalisation model — are planned and described here as the target system design, to be implemented in subsequent development stages.

D. Memory and Personal Context

Long-term personalisation requires memory, but memory in LifeXA is practical rather than unlimited. Stored items include preferences (for example, concise reminders), routines (usually studies after 8 PM), task context (project deadline on Friday), communication preferences (direct wording) and explicitly saved information. Explicit memories are stored only with clear user intent, and all stored items can be corrected or deleted.

E. Adaptive Reminder Logic

A conventional reminder reads “Assignment — 7:00 PM.” A LifeXA-style message instead reads, in essence: “Your assignment is due tomorrow. You usually study after 8 PM and have about 45 minutes available tonight. Would you like me to schedule a 45-minute study session?” The second message uses context and preference, and asks for confirmation instead of changing the schedule silently, preserving autonomy.

Fig. 4 summarises the decision logic. When a deadline approaches, the system retrieves routine, preferences and calendar context. If the task has been postponed repeatedly, it offers a smaller starting step or rescheduling; otherwise it checks for a free slot in the preferred window, sending a reminder if one exists and proposing an alternative if not. Changes are made only after the user accepts, and the outcome, including overrides, updates memory.

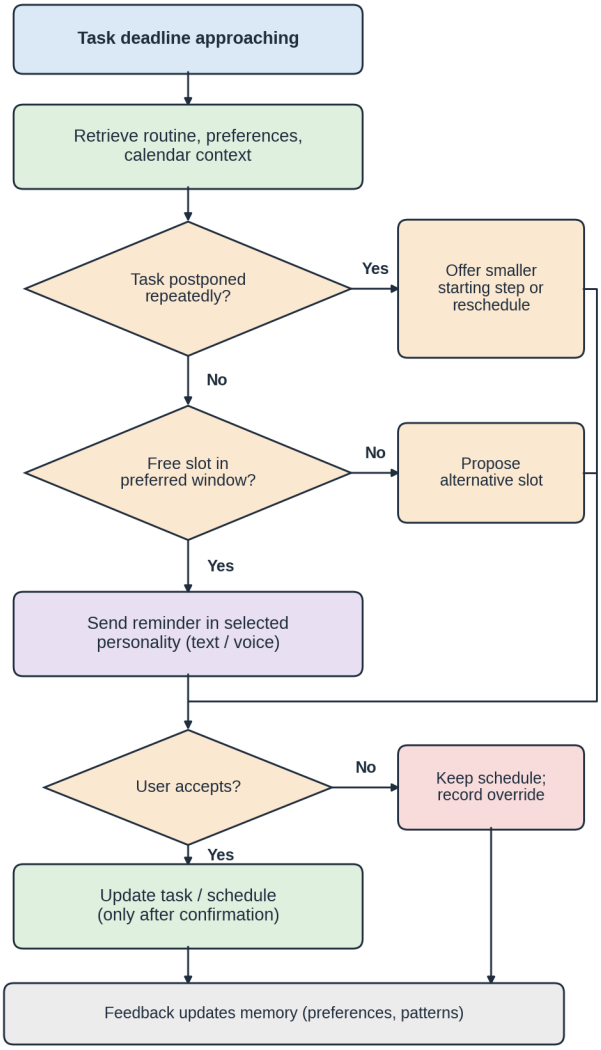


Fig. 4. Adaptive reminder decision flowchart with user confirmation and memory feedback.

F. Voice and Selectable Personality

Voice makes the assistant easier to use where typing is inconvenient and brings it closer to an intelligent personal assistant than a conventional productivity app. Personality means communication style, not a fake human identity. Table II lists the planned styles. The personality changes tone, wording and voice style while the same safety, privacy and task-management rules apply, so a professional and an energetic assistant may phrase one reminder differently, but neither invents information or acts outside the user’s permission.

TABLE II
SELECTABLE PERSONALITY STYLES

Style	Communication behaviour
Professional	Clear, formal and focused
Friendly	Warm but still task-oriented
Calm	Gentle, simple and less intense
Energetic	Positive and action-focused

Style	Communication behaviour
Minimal	Short answers with only necessary information
Detailed	More explanation and context

VI. RESEARCH METHODOLOGY AND EVALUATION PLAN

A. Staged Development

Development proceeds in eight stages so that the research does not depend only on the final application (Fig. 5). Each stage produces something testable: literature review, requirement study, base assistant, LLM integration, personalisation and memory, adaptive reminders, voice and personality, and a final evaluation.

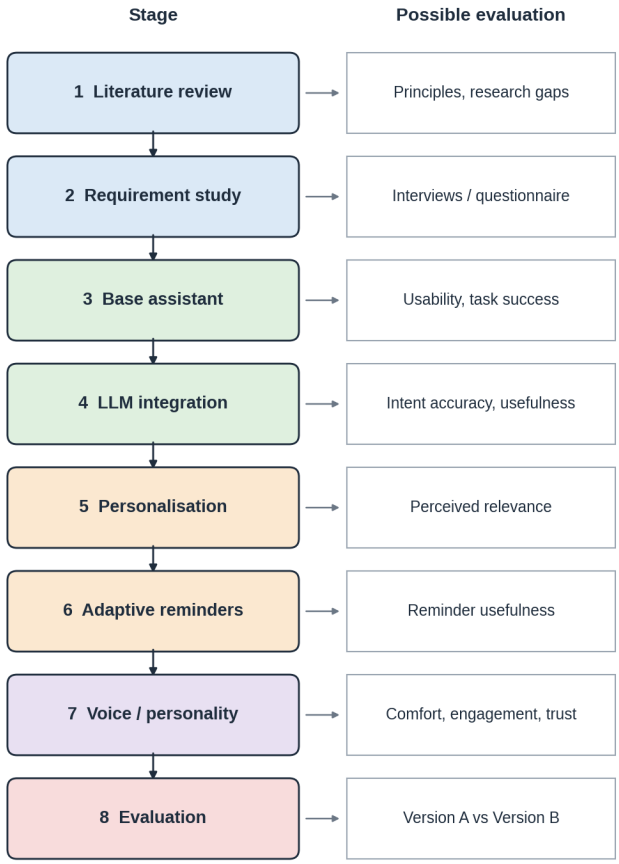


Fig. 5. Staged research methodology and the evaluation associated with each stage.

B. Comparative Design

A useful design compares two versions. **Version A** is a conventional assistant with fixed reminders, limited personalisation and a standard response style. **Version B** is the LifeXA personalised assistant with user preferences, relevant memory, adaptive reminders, LLM interaction and a selected personality. The exact experimental design, sample size and statistical analysis will be decided after the literature review and with faculty guidance.

C. Measures

Table III lists the planned measures. They will be complemented by qualitative questions on whether the assistant understood the user's intent, whether reminders arrived at useful times, whether suggestions felt relevant or intrusive, whether personality and voice helped, and whether users felt in control of their schedule.

TABLE III
PLANNED EVALUATION MEASURES

Area	Possible measure
Task management	Tasks completed, postponed or rescheduled
Reminder usefulness	User rating of timing, wording and relevance
Perceived usefulness	Whether the assistant actually helps
Ease of use	How easy the system feels to operate
Engagement	Frequency and consistency of use
Personalisation	Whether suggestions feel relevant to the individual
Voice experience	Comfort and usefulness of spoken interaction
Personality preference	Which styles users prefer and why
Trust	Confidence in responses and actions
User autonomy	Whether users feel in control of decisions

D. Working Hypotheses

The following hypotheses may be refined after the literature review and a pilot study. **H1:** users of the personalised version will report higher perceived usefulness than users of a conventional reminder-based assistant. **H2:** adaptive reminder timing will be perceived as more relevant than fixed timing. **H3:** users who select a communication style will report higher interaction satisfaction than users given a fixed style. **H4:** relevant memory and preference-based personalisation will improve perceived relevance, provided users remain in control of stored information. **H5:** voice interaction will improve naturalness and convenience for at least some users.

VII. ETHICS, PRIVACY AND USER AUTONOMY

Because LifeXA stores personal information and may learn behavioural patterns, privacy is part of the system design rather than an afterthought. The system should collect only information useful for its stated purpose, give users control over stored memories and preferences, and allow them to correct or delete information. It should clearly distinguish generated suggestions from confirmed actions, ask before making meaningful changes to schedules or tasks, avoid unnecessary collection of sensitive information, and use appropriate security for stored data and API communication. Personalised features should be shown to be helpful before they are made automatic.

The assistant's role must stay clear: it is a personal assistant, not a therapist, and it must not encourage emotional dependency or exclusivity. This boundary matters because human-like voice and personality can make an AI system feel more social, and anthropomorphism and relationship-like perceptions can influence trust and interaction [6]. The design therefore balances natural interaction with clear system boundaries.

VIII. EXPECTED LIMITATIONS

A small student study may not represent all types of users, and preferences and routines change over time, so past behaviour may not predict future behaviour. LLM responses can be incorrect or inconsistent. Voice quality and speech recognition can affect the experience independently of psychology, and personalisation can become annoying if it is too frequent or inaccurate. It may be hard to separate the effect of personalisation from the effect of a better interface, and long-term behavioural effects may need a longer study than the project timeline allows.

IX. EXPECTED CONTRIBUTIONS AND FUTURE SCOPE

LifeXA is not simply another productivity application. It aims to show how psychological ideas can be translated into practical assistant features: a framework for psychologically informed assistance, a hybrid LLM and rule-based architecture, a personalisation model based on preferences, routines, behaviour and context, adaptive reminder logic, voice with selectable personalities, user-controlled memory, an evaluation approach for usefulness, engagement, trust and autonomy, and a clear boundary between a personal assistant and a therapeutic or companion system.

Future work includes context-aware assistance using calendar, location or device context where consented; better prediction of suitable task times; personalised task breakdown; more natural voice conversation and more configurable personality dimensions; multilingual interaction; improved memory retrieval and correction; privacy-preserving or on-device processing where feasible; and longer studies comparing fixed and personalised reminder strategies.

X. CONCLUSION

LifeXA is designed as a psychologically informed AI personal assistant that knows not only what the user has asked, but also enough about the user's preferences, routines and context to give more relevant help. The LLM makes interaction flexible and conversational, voice makes it more accessible, personality customisation lets users decide how the assistant communicates, and memory provides continuity, while psychological principles guide reminders, task breakdown and scheduling. LifeXA remains an assistant: it helps the user organise and act, and the user stays in control of important decisions, stored information and the assistant's behaviour. The next step is to build the prototype and carry out the planned comparative study to test whether an assistant designed around real human behaviour is more useful than one that treats every user and task the same way.

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